Contents

AI-Powered Clinical Documentation Improvement System

** :** - (SAIP) -

: 2025

/

** :** (CDI System)

: |

** (IPC):** - G16H 10/60 ( ) - G16H 50/20 ( ) - G06N 5/02 ( )

:

" ICD-10-CM"

:

    (Google Gemini 2.0)   (ICD-10-CM)   .

:

1. ** **: 40-60%
2. ** **: 15-25%
3. ** **: 15-30
4. ** **: CDI specialists

:

    30   95% 60%   40%.

1.

: .

** :**

```python
def analyze_clinical_note_advanced(note_text, patient_context):
    """


    Input:
    - note_text:
    - patient_context:   (  )

    Process:
    1.    (NLP)
    2.
    3.
    4.
    5.
    6.   ICD-10-CM
    7.
```

```
    Output:
    - gaps:
    - icd10_codes:
    - physician_queries:
    - cdi_notes:
    """
    #
    pass
```

** :** - (/) -
-
- (>95%)

2. MFA

: .

: - OTP
- 30 ( ) -
-

3.

: .

** :**

```
//
const ChatSystem = {
  types: [
    {
      name: "  ",
      questions: [
        "      ",
        "   ICD-10-CM  ",
        "   ",
        // 12+
      ]
    },
    {
      name: " ",
      features: [
        "  ",
        "  ",
        "  "
      ]
    }
  ],
  specialties: [
    " ",
    "",
    " ",
    // 20+
  ]
}
```

4.

: .

```python
class MedicalDataEncryption:
    """

    """

    #  1:
    @staticmethod
    def encrypt_password(password: str) -> str:
        """bcrypt with adaptive cost factor"""
        return bcrypt.hashpw(password, bcrypt.gensalt(rounds=12))

    #  2:
    @staticmethod
    def encrypt_phi(data: str, key: bytes) -> str:
        """AES-256 for PHI (Protected Health Information)"""
        f = Fernet(key)
        return f.encrypt(data.encode()).decode()

    #  3:
    # TLS 1.3 for all communications

    #  4:
    # MongoDB encryption at rest
```

5.

: .

```
Admin ()
```

```
Supervisor ( )
```

```
User (/)
```

```
      AI
```

6. API Key Rotation

: API .

```python
class APIKeyRotation:
    """

    """

    def __init__(self, keys: List[str]):
        self.keys = keys
        self.usage = {key: 0 for key in keys}

    def get_key(self) -> str:
        """
```

```python
        """
        min_key = min(self.usage, key=self.usage.get)
        self.usage[min_key] += 1
        return min_key

    def report_error(self, key: str):
        """

        """
        self.keys.remove(key)
        #   cooldown period
```

1.

- 40%
- 15-25%
- 60%

2.

-
-

3.

-
-

4.

-
-

:

****	+	
** **	< 30	15-30
****	>95%	70-85%
****		
****		
****	MFA +	
** **	Gemini 2.0	
** **		

---

1. (Patent)

** :**

** 1 (Claims):** : -
-
- ICD-10-CM -
-

** 2:** : 1.
2.
3.
4.
5.

** 3:** : - (MFA) - AES-256 -
-

2. (Copyright)

** :** - (Frontend + Backend) - UX/UI Design -
-
-
-

3. (Trademark)

** :** - : "CDI Smart System" - (Logo) -
-

4. (Trade Secrets)

** :**

-
- AI
-

1.

```
Backend (Python/FastAPI)
   server.py (1800+ lines)
   security_*.py (800+ lines)
   requirements.txt
Frontend (React.js)
   src/
      pages/ (15 pages)
      components/ (30+ components)
      utils/
   package.json
Database
    MongoDB schemas
```

2.

-
-
- API Documentation

3. (Screenshots)

-
-
-
-

4.

- (>95%)
- (< 30)
-
-

5.

- HIPAA Compliance
- ISO 27001
- OWASP Top 10

:

1. ** ():**
    - 500+
    - : 200-300
2. ** ():**
    - 2000+
    - : 800 - 1.2
3.

    **** :****

    - 22
    - : 5+

:

1. **SaaS Subscription:**
    - : 5,000 - 20,000
    - : 50,000 - 200,000
2.

Enterprise Licensing:

-

3.

API Access:

- Pay-per-use model

()

1.
2.
3.
4.

** :** - Email: ip@cdi-system.sa - Phone: +966 XX XXX XXXX

** ( ):** - : - : - :

/ .

: : :

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1. Clinical Documentation Improvement Best Practices (AHIMA)
2. ICD-10-CM Official Guidelines (WHO)
3. HIPAA Security Rule (HHS)
4. AI in Healthcare: Current Applications (Nature Medicine)
5. Natural Language Processing in Clinical Documentation (JAMIA)
